Vehicle Parking Management and Tracking System

Software Requirements Specification (SRS)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-01	Functional	The system shall allow users to register an account and log in using valid credentials.	High	A user can register with valid details and log in; invalid credentials are rejected with an error message.	Every protected function needs a known, authenticated user.	UC-01. Source: Driver.
FR-02	Functional	The system shall allow users to register, view, update and manage their vehicle information: registration number, vehicle type, model and colour.	High	A vehicle is saved and listed; a duplicate or invalid registration number is rejected; edits persist.	Entry, tracking and history depend on registered vehicles.	UC-02. Source: Driver.
FR-03	Functional	The system shall allow users and authorised parking staff to view the current availability and status of parking slots: Available, Occupied, Reserved or Unavailable.	High	Slots show the correct status, can be filtered by status, and update after every entry, exit or reservation.	Drivers and attendants need to know which slots are free.	UC-03. Source: Driver, Parking Attendant.
FR-04	Functional	The system shall allow authenticated users to reserve an available parking slot for a specified period.	High	A reservation is created for an available slot and the slot shows Reserved; a taken slot or an end time before the start time is rejected.	Lets drivers secure a slot before they arrive.	UC-04. A reserved slot cannot go to another vehicle (DR-08).
FR-05	Functional	The system shall allow authorised parking staff to record vehicle entry and create an active parking session.	High	Entry creates a session with the correct entry time; an unregistered vehicle or a vehicle that already has an active session is rejected.	The parking session is the basis for tracking, fees and history.	UC-05. One active session per vehicle (DR-02).
FR-06	Functional	The system shall assign an appropriate available parking slot to a vehicle and prevent duplicate slot assignments.	High	A compatible free slot is assigned and shows Occupied; an occupied, reserved or unavailable slot is never assigned.	Prevents two vehicles being sent to the same slot.	UC-06. Vehicle type must match the slot type (DR-07).
FR-07	Functional	The system shall allow authorised users to identify the current parking location and status of a parked vehicle.	High	Searching a parked vehicle returns its slot and status; a vehicle that is not parked returns "no active session".	Helps staff locate parked vehicles quickly.	UC-07. Parking Attendant and Administrator only.
FR-08	Functional	The system shall allow authorised parking staff to record vehicle exit and complete the parking session.	High	Exit records the time, triggers fee calculation and releases the slot; an exit with no active session, or before the entry time, is rejected.	Closes the session and frees the slot for the next vehicle.	UC-08. DR-04, DR-09.
FR-09	Functional	The system shall calculate parking duration and charges according to the configured parking rates.	High	Fee = duration x configured rate (2 hours at Rs 20 per hour = Rs 40); a changed rate applies to new calculations.	Removes manual fee errors.	UC-09. Rates are set by the Administrator.

FR-10	Functional	The system shall allow users or authorised parking staff to make and record parking payments.	High	A successful payment marks the session Paid; a failed payment is recorded and the user is told; each payment links to one session.	Keeps a reliable payment record for every session.	UC-10. Uses the external Payment Gateway.
FR-11	Functional	The system shall maintain and display historical parking-session information.	Medium	History lists past sessions with vehicle, slot, entry and exit times, duration, fee and payment status; past sessions cannot be edited.	Drivers and administrators can review past parking.	UC-11. Source: Driver, Administrator.
FR-12	Functional	The system shall allow administrators to manage users, vehicles, parking slots and parking rates, and generate parking reports.	High	An administrator can add or remove users, add or disable slots, change rates and generate an occupancy report.	Gives the facility central control and reporting.	UC-12. Source: Administrator.
NFR-01	Non-Functional	The system shall display parking-slot availability within 2 seconds for at least 95% of requests under normal operating conditions.	High	In a load test, 95% of availability requests return within 2 seconds.	Availability is the most frequent request.	Performance.
NFR-02	Non-Functional	The system shall process normal vehicle-entry and vehicle-exit transactions within 3 seconds, excluding external payment-processing time.	High	Entry and exit complete within 3 seconds in testing, not counting payment time.	Avoids queues at the gate.	Response time.
NFR-03	Non-Functional	The system shall maintain at least 99% availability during defined operating hours.	Medium	Measured uptime during operating hours is at least 99%.	The parking facility runs continuously.	Availability.
NFR-04	Non-Functional	The system shall support at least 500 parking slots, 5,000 registered vehicles and 100 concurrent users.	Medium	The system works correctly when tested with 500 slots, 5,000 vehicles and 100 concurrent users.	Sized for a large facility.	Scalability.
NFR-05	Non-Functional	A parking attendant shall be able to complete a normal vehicle-entry transaction within 1 minute after basic system training.	Low	A trained attendant completes a normal entry within 1 minute.	Keeps entry fast for staff and drivers.	Usability.
NFR-06	Non-Functional	The system shall keep parking-slot and parking-session information consistent when several users perform parking operations at the same time.	Medium	Simultaneous entries never assign one slot twice or create two sessions for one vehicle.	Prevents double assignment under load.	Reliability.
NFR-07	Non-Functional	The system shall be built from modular components for user management, vehicle management, parking management, payment and reporting.	Medium	The code has a separate module for each of the five areas.	Makes the system easier to change and test.	Maintainability.
NFR-08	Non-Functional	The system shall support common web browsers: Chrome, Edge and Firefox.	Low	All main flows work in Chrome, Edge and Firefox.	Users have different browsers.	Portability.
NFR-09	Non-Functional	The system shall keep accurate and consistent relationships between users, vehicles, parking slots, parking sessions and payments.	Medium	Every session links to one vehicle and one slot, and every payment links to one session.	Wrong links give wrong fees and history.	Data integrity.
SEC-01	Security	The system shall require users to authenticate before accessing protected functions.	High	Protected pages and requests are refused without a login.	Protects user and payment data.	Authentication.
SEC-02	Security	The system shall give different access privileges to Drivers, Parking Attendants and Administrators.	High	An action outside the user's role is denied, for example a driver trying to track a vehicle.	Each role sees only what it needs.	Role-based authorisation.

SEC-03	Security	The system shall store passwords using a secure password-hashing mechanism.	High	No plain-text passwords are stored; a recognised hashing algorithm such as bcrypt is used.	Limits the damage if data leaks.	Password security.
SEC-04	Security	The system shall temporarily lock a user account after 5 consecutive unsuccessful login attempts.	High	The fifth failed login in a row locks the account and shows a message.	Stops password guessing.	Failed-login protection.
SEC-05	Security	The system shall automatically end inactive authenticated sessions after 15 minutes.	High	After 15 minutes of inactivity the user is sent back to the login page.	Protects unattended devices.	Session security.
SEC-06	Security	The system shall validate user input before processing or storing it.	High	Invalid input is rejected with a clear message.	Prevents bad data and common attacks.	Input validation.
SEC-07	Security	The system shall keep audit logs of important administrative activities.	High	Key admin actions are logged with the action, the admin and the time.	Makes admin changes traceable.	Audit logging.
SEC-08	Security	Only authorised administrators shall be able to access audit logs.	High	Users who are not administrators are denied access to audit logs.	Protects the audit trail.	Audit access control.
SEC-09	Security	The system shall prevent unauthorised users from accessing confidential user and payment information.	High	Users cannot see other users' details or payments; card details are never stored.	Protects personal and payment data.	Data protection.